



South Africa liquid fuels market story

Inland supply, corridor capture and tank
turnover

Durban-Lesedi corridor into Gauteng

8 October 2026

Reported Vopak context and illustrative investment economics; all
commercial assumptions require validation

Photo source: Royal Vopak | Terminal Durban | vopak.com

Strictly Confidential



Slow demand growth and greater import reliance set the baseline; road-to-rail shifts, EV adoption, power recovery and refinery changes could reshape the outlook

Baseline / reference	What the evidence establishes	What changes the outlook
 Fuel demand and power	2023 Department sales: 21.94bn L. Petrol CAGR -2.08%; diesel +0.82%. Own Eskom fuel fell 40% in FY2025.	Sector activity, fleet replacement and power dispatch change fuel use. 2024 sales remain unverified.
 Domestic supply and imports	2020–2025 imports rose by 5.43bn litres of diesel and 2.73bn of petrol. The sourcing mix has shifted.	SAPREF and PetroSA restart paths, Secunda gas/MRG and Natref output can change domestic supply.
 Corridor position and tank turnover	Durban and Lesedi provide coastal and inland assets serving Gauteng. Working capacity and turns are open.	Route costs, access and customer capture determine flow; achievable turnover determines handling headroom.
 Medium / Medium reference world	M/M anchors the nine-world matrix. Historical evidence is assembled; numerical calibration remains open.	Vary demand and supply around M/M, then test capture, turnover and returns to identify investment triggers.

Department sales 2013–2023; 2023 total 21.94bn L. FIASA 2024 total 20.76bn L is unverified; no 2025 sales. SARS imports 2020–2025. Eskom FY2025 own diesel + kerosene. M/M remains an uncalibrated reference.

Prioritise capture and turnover through existing assets; expand where profitable demand exceeds achievable capacity

Working recommendation	Investment logic	Condition for commitment
 Capture inland supply	Win accessible customer flows from imported and domestic sources.	Prove delivered competitiveness and sufficient customer volume.
 Use existing assets first	The worked illustration favours existing assets in six worlds; access/dispatch improvement in two.	Confirm working capacity and the actual operating constraint.
 Stage corridor and capacity spend	Only high demand / low domestic supply favours extra tanks, with a narrow advantage over improvement alone.	Fund the least-cost viable option that removes the proven constraint.
 Keep strategic options open	Test all nine worlds and emerging gateways; screen EV adjacencies separately.	Validate capture, chargeable revenues and capex first; the illustrated choice reverses under downside tests.

Working recommendation supported by an authored Lesedi illustration, not a calibrated forecast. Reported capacity is used; capture, working capacity, tariffs, capex and discount rate are assumptions. No customer volume is represented as secured.

The illustrated expansion case passes narrowly—and depends on capture, revenue and capex

Decision test	Numeric condition	ILLUSTRATIVE high / low case	Still needs verification
 Customer volume	Tanks must outperform improving existing capacity.	Assumed capture 2.813bn L/year; tanks win above 2.806bn L/year.	No secured volumes inferred. Validate customer capture and contract duration.
 Route economics	Delivered-cost headroom $\geq$ R0/m ³ .	R750 allowable less R650 assumed logistics = R100/m ³ headroom.	Both prices are authored. Compare actual Durban, Maputo, Walvis Bay and domestic quotes.
 Capacity constraint	Captured flow exceeds achievable handling.	Existing 1.890bn L/year; improvement 2.268bn; tanks 2.916bn.	90% working capacity and 1.25 / 1.50 monthly turns are assumptions, not reported operations.
 Incremental return	Positive NPV and better value than the alternative.	Improvement NPV R+27m; combined R+35m. Extra advantage only R8m.	10% discount rate, 20 years, pre-tax. Actual tariffs, cost, tax and project schedule remain open.

ILLUSTRATIVE ONLY. 100% project basis, constant ZAR; capex at t0; operating ramp 50%/75%/100%; no tax, working capital, construction delay, inflation or residual value. Turnover and occupancy are distinct. Full assumptions, benchmarks, options and downside tests: pp39–42.

Only one illustrative world favours tanks; six favour existing assets and two favour improvement

Gauteng demand / domestic supply	HIGH deliveries 4.8bn L/year	MEDIUM deliveries 3.6bn L/year	LOW deliveries 2.4bn L/year
LOW demand 6.0bn L/year	L / H USE EXISTING ASSETS Capture 0.99bn L/year Incremental NPV R+0m	L / M USE EXISTING ASSETS Capture 1.50bn L/year Incremental NPV R+0m	L / L USE EXISTING ASSETS Capture 2.01bn L/year Incremental NPV R+0m
MEDIUM demand 6.8bn L/year	M / H USE EXISTING ASSETS Capture 1.39bn L/year Incremental NPV R+0m	M / M BASELINE USE EXISTING ASSETS Capture 1.90bn L/year Incremental NPV R+0m	M / L IMPROVE ACCESS / DISPATCH Capture 2.41bn L/year Incremental NPV R+27m
HIGH demand 7.6bn L/year	H / H USE EXISTING ASSETS Capture 1.80bn L/year Incremental NPV R+0m	H / M IMPROVE ACCESS / DISPATCH Capture 2.30bn L/year Incremental NPV R+27m	H / L TEST IMPROVEMENT + TANKS Capture 2.81bn L/year Incremental NPV R+35m

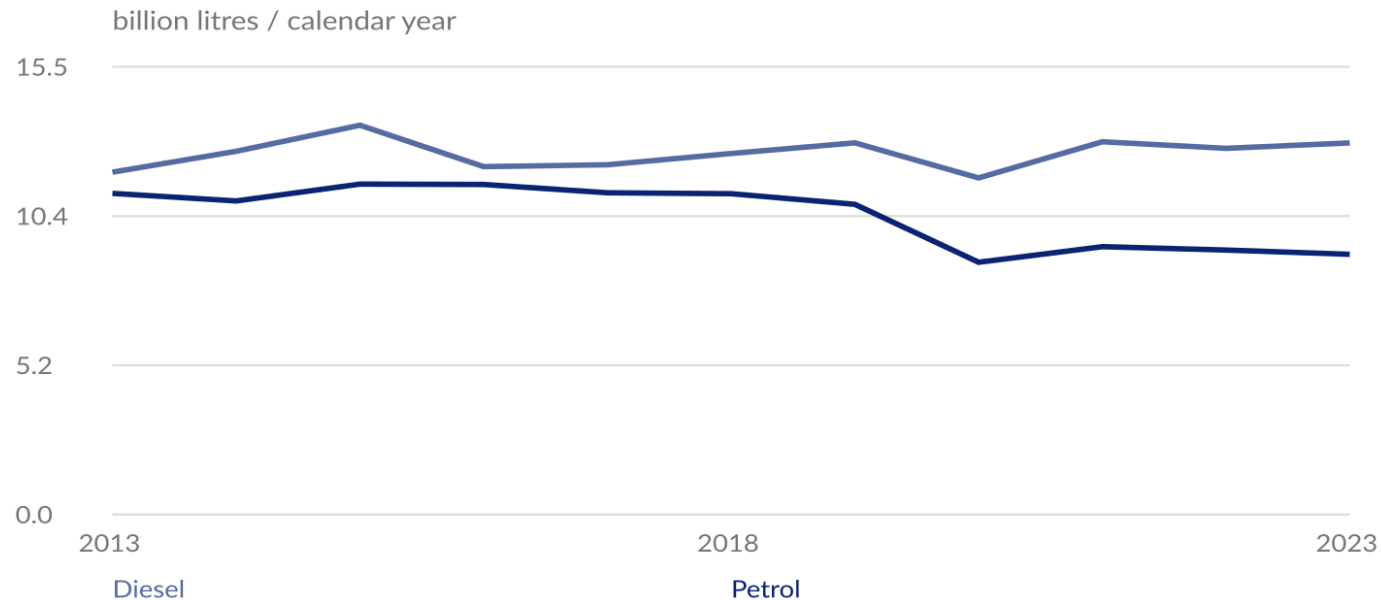
Highest incremental NPV among three illustrative packages; R0 means retain existing assets, not zero terminal value.

ILLUSTRATIVE snapshot, not forecast or probabilities. Domestic supply here means assumed deliveries into Gauteng, not national refinery capacity. Shares, rates and costs held fixed across worlds. Rail/EV opportunities remain separate cases, not valued here. Each cell is relative to existing assets; unserved volumes can remain. Inputs: p41; downside: p42.

Petrol demand has declined while diesel has grown slowly, limiting the case for demand-led expansion outlook

Demand trend

Change over 2013–2023



01 | Petrol 2013–2023

11.15 to 9.04 bn litres
-2.12 bn L / -19.0%
CAGR -2.1% per year

02 | Diesel 2013–2023

11.89 to 12.91 bn litres
+1.02 bn L / +8.6%
CAGR +0.8% per year

FIASA 2024 observations are excluded from this comparison: source definitions are not reconciled.

2023 Department sales total 21.94bn L. The 2024 FIASA comparison is 20.76bn L, unverified; no 2025 sales figure is available.

Department sales 2013–2023: petrol CAGR -2.08%, diesel +0.82%. FIASA 2024 is excluded from the trend. Workbook History Q71/R71. RAF levy evidence uses fiscal years and still needs reconciliation.

Power shortages and road freight have supported diesel consumption, but that support can change with increased support for rail reform and improved Eskom

Demand support

2024 vs 2025

■ 2024 ■ 2025

Implication for demand

Power generation

Own OCGT fuel • million litres • financial years

OCGT fuel



-39.9%

450m fewer litres in one year. Sustained recovery reduces diesel demand; retirements and replacement timing could reverse it.

EAF: 54.56% → 60.60% | Cost*: R23.87bn → R13.32bn

Road freight

Reported payload • million tonnes • calendar years

Road



-0.3%

Rail



+4.7%

Road carries 85.3% of tonnes. Workbook truck-fuel proxy: 3.15bn L in 2023, partial operator coverage.

Sector activity

Real GVA • Rbn, 2015 prices • calendar years

Agriculture



+17.4%

Mining



+0.2%

Manufacturing



-1.2%

Activity differs by sector. 2024 indicative diesel: mining 1.40, industry residual 0.17, farms 0.96bn L.

Eskom Integrated Report 2025 pp100,141: own-fleet diesel + kerosene; *cost includes storage/demurrage. Stats SA P7162 Dec 2025 p7: both years from the revised vintage. Stats SA P0441 Q2 2026 annual real GVA: agriculture includes forestry/fishing. FY and calendar-year periods shown separately. Bars start at zero; scales are common within each panel, not across different units. Light blue = 2024; navy = 2025. Workbook Sector history: post-2021 litres estimated. Freight proxy = survey fuel spending / wholesale price; partial coverage.

Mining, manufacturing and agriculture link diesel demand to sector activity

Sector	2019 diesel bn litres	2021 diesel bn litres	2024 estimate bn litres	Activity and interpretation
Mining	1.67	1.29	1.40	2024 production index 93.5 (2019=100). Fuel intensity varies materially; mine and on-road use overlap.
Manufacturing / non-specified industry	0.23	0.16	0.17	2024 output index 93.3. Fuel is industry less mining/construction, not all industrial use.
Agriculture / forestry	0.92	1.06	0.96	2024 GVA R114.8bn, 2015 prices. Weather, crop cycles, irrigation and transport affect exposure.

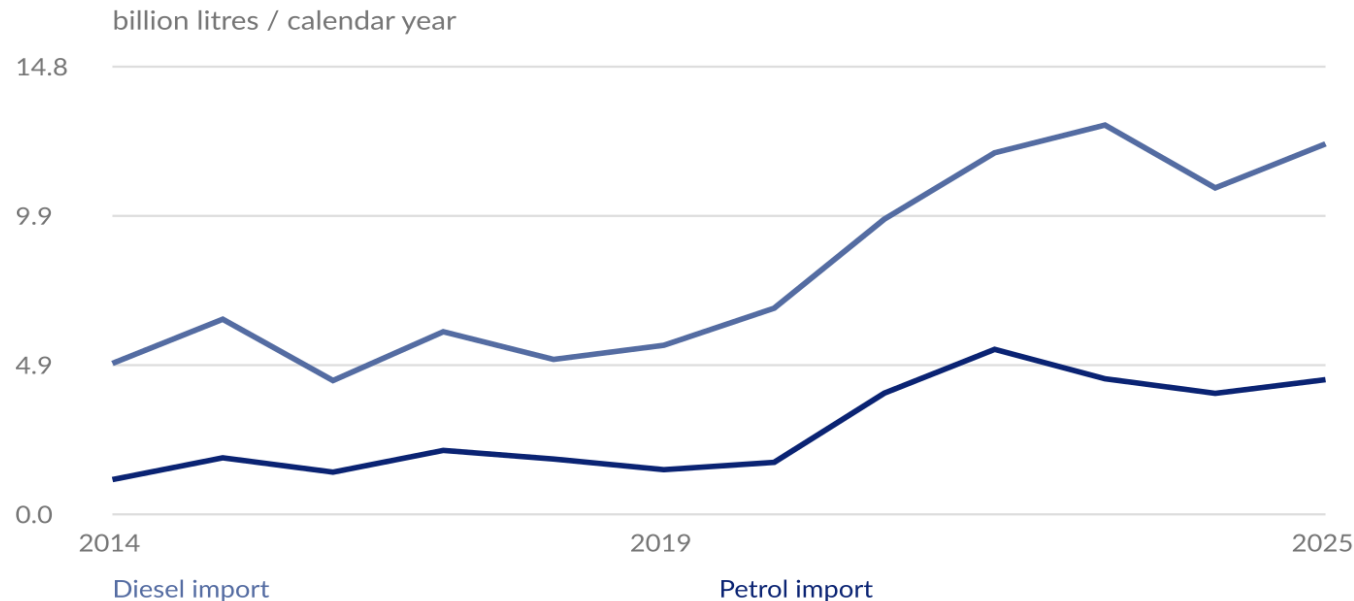
Post-2021 fuel estimates hold 2018-2021 average intensity. Mining/manufacturing use litres per index point; agriculture uses litres per real rand of activity.

Workbook Sector history M6/O6/R9, M16/O16/R19, M22/O22/R25. Department energy balance to 2021; later fuel estimated with activity and average intensity. Sector attribution and road overlap require review.

The shut down of domestic refining capacity has lowered domestic production and has increased imports shifting how inland demand is supplied

Import trend

Change since 2020



01 | Diesel import

6.81 to 12.25 bn litres
Increase: 5.43 bn litres
CAGR: 12.4% (2020–2025)

02 | Petrol import

1.72 to 4.45 bn litres
Increase: 2.73 bn litres
CAGR: 21.0% (2020–2025)

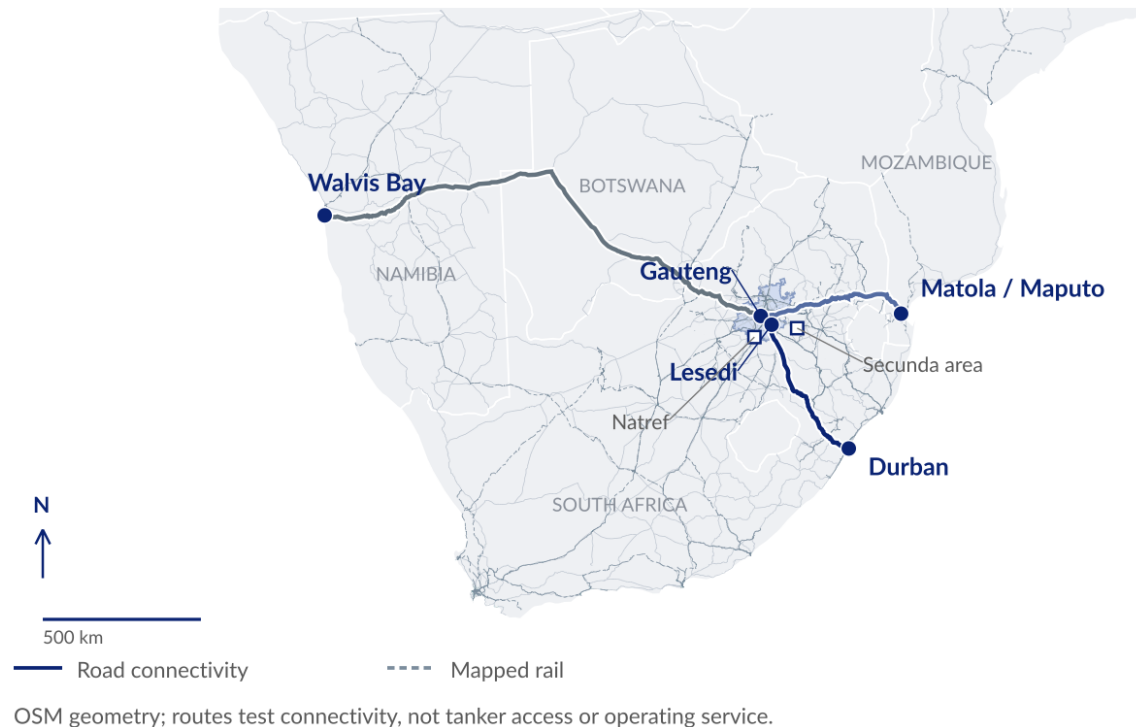
More inland handling opportunity;
capture and turnover determine benefit.

2025 gross imports: diesel 12.25bn L and petrol 4.45bn L. Combined net imports: 15.15bn L; port allocation and Vopak capture remain open.

SARS litre-reported petrol/diesel codes, complete calendar years. 2020–2025 changes are historical comparisons, not plant-level causal attribution. Export histories retained in the source report; unit/blend exclusions remain.

Emerging Maputo and Walvis Bay routes must be tested alongside Durban and domestic supply

Competing supply corridors



Same product, same destination

01 | Compare delivered cost

Product, handling, transport, border, losses and inventory finance.

02 | Establish price headroom

Bearable charge less actual logistics cost. Capacity and reliability constrain capture.

03 | Include domestic sources

Secunda and Natref also serve inland demand; routes and access need validation.

Next action

Match route costs and access to the same product and Gauteng destination.

Maputo and Walvis Bay are emerging alternatives to include by destination. Compare delivered cost, capacity and reliable access before estimating how much inland demand Vopak can capture.

© OpenStreetMap contributors (ODbL; osm.org/copyright) / Geofabrik; Natural Earth; geoBoundaries. Shared LAEA projection. Paths test road connectivity, not tanker access. Secunda uses a city reference; Natref a mapped site. Terminal access, turn restrictions and operating rail/pipeline services remain unvalidated. Emerging alternatives are in scope; relative competitiveness is not yet quantified.

Durban and Lesedi connect coastal supply to Gauteng; operating data must establish room for growth

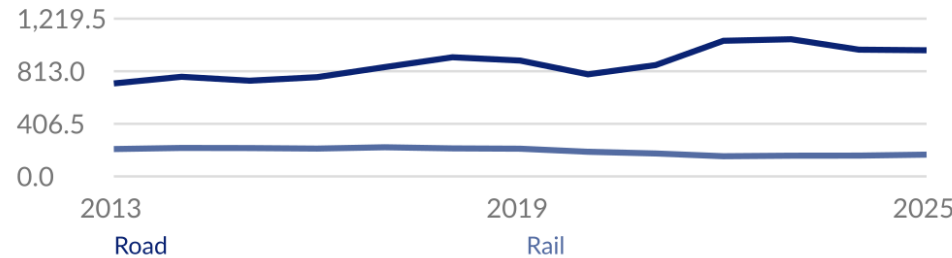
Asset / measure	Established position	What must be established
Durban	360,246 m ³ published gross capacity. Coastal receipt and onward distribution; mixed products.	Isolate petrol/diesel working tankage and actual throughput by product.
Lesedi	140,000 m ³ clean-product capacity. Includes the 40,000 m ³ expansion commissioned in October 2025.	Confirm usable working capacity, annual deliveries and monthly tank turns.
Operating constraints	Storage capacity establishes the asset footprint. It does not establish annual handling capacity.	Measure receipt, pipeline and dispatch limits, seasonal peaks and stock requirements.
Implication for growth	Actual throughput and achievable turns are not yet established in this review.	No spare capacity is claimed. Test captured demand against verified handling limits before selecting investment.

Sources: Vopak terminal directory; Annual Report 2025 confirms Lesedi expansion commissioned in 2025. Published capacity is not working capacity or throughput. Operating data request: p18. Illustrative turnover sensitivity moved to p43; it is not observed utilisation.

Accelerated rail recovery, EV adoption and ICE technology efficiency could reduce transport fuel demand in the medium to long term

Rail recovery

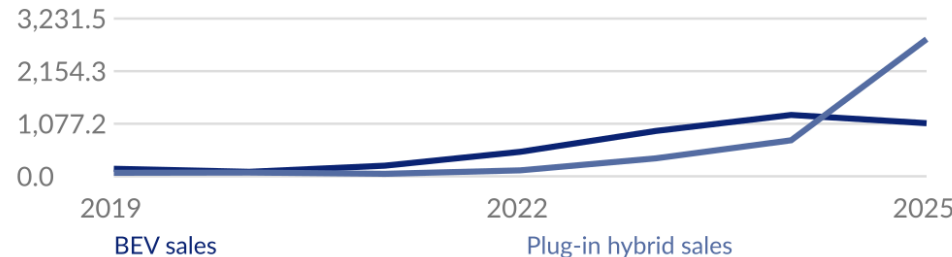
million tonnes



D2: capacity, reliability, transferable freight
Tonnes require haul distance and traction
before conversion to net diesel savings.

EV uptake

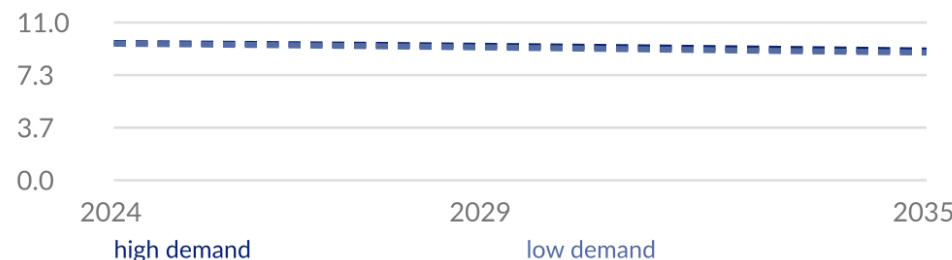
vehicles sold / year



D3: affordability, charging, fleet replacement
Sales history is not fleet penetration.
L/M/H adoption calibration remains open.

Fleet efficiency

L/100 km; draft engine

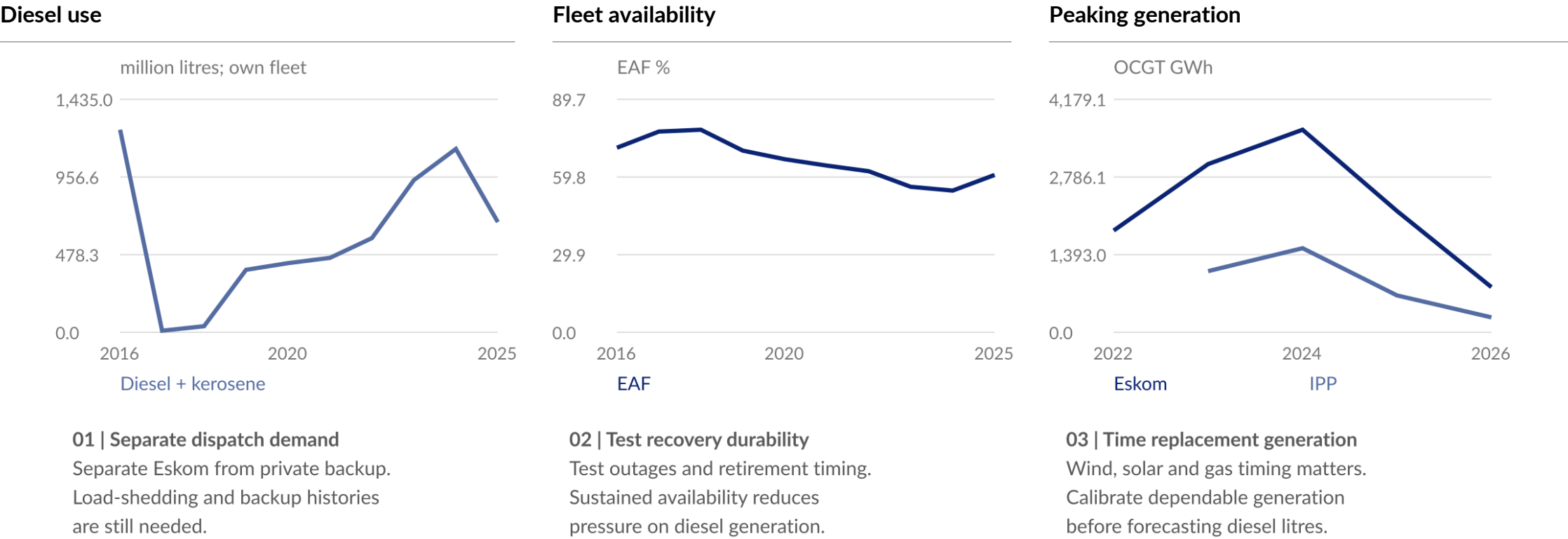


D4: new-cohort efficiency and mileage
Dashed paths are existing model diagnostics.
OEM history and Medium calibration are open.

2025 sales: 1,088 BEVs (0.18% of all new sales), 2,810 plug-in hybrids and 12,818 conventional hybrids. Each has a different fuel effect.

Stats SA freight; naamsa all-market sales 2019-2025, Workbook Vehicle history S31:S36. BEV/PHEV/HEV sales are distinct from fleet shares. Efficiency paths remain model assumptions, with OEM history and annual calibration open.

Power generation recovery could lower diesel use further, while outages and replacement delays could sustain domestic demand



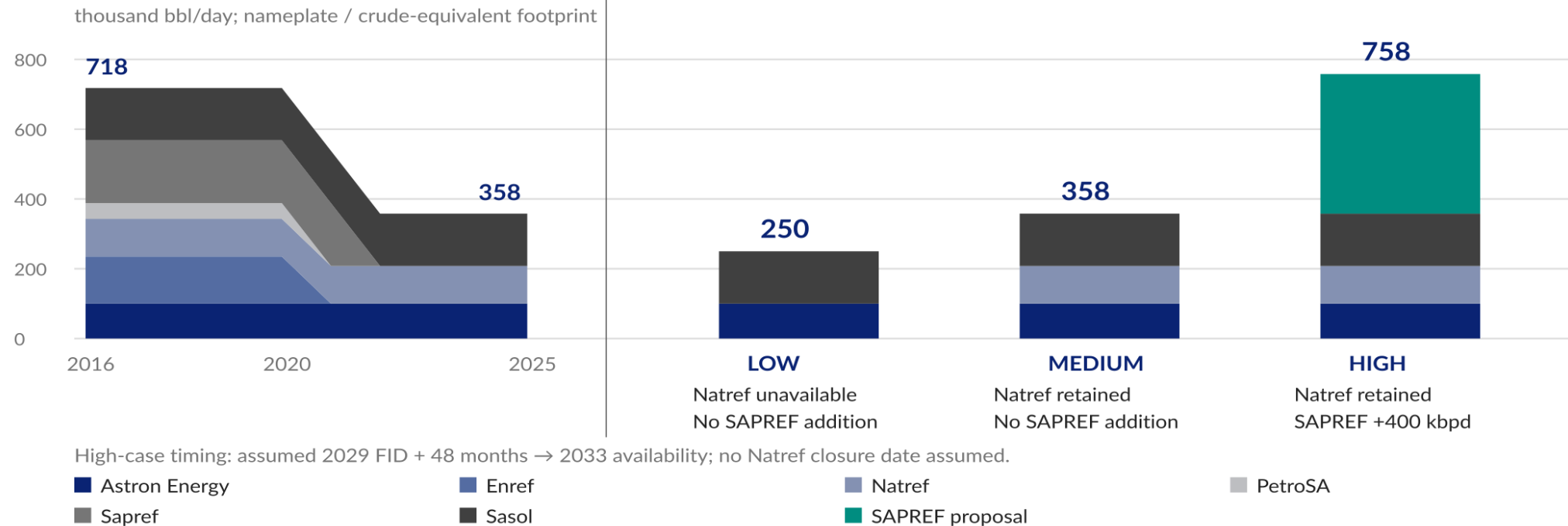
Own Eskom fuel is reported separately. The workbook adds Eskom/IPP generation and four diesel stations; litres derived from output and future dispatch remain estimates.

Eskom FY ending March: reported own diesel + kerosene; Eskom/IPP GWh include kerosene output. Workbook Power fleet separates the four diesel stations. Gas switch, coal-site repowering and L/M/H settings await Nigel/Henry review.

SAPREF could expand the supply footprint, while Natref unavailability would reduce it

Reported history

Illustrative 2036 capacity cases



These capacity cases anchor the supply-axis discussion. Translate them into product output, then test imports, corridor capture and required tank turns in each of the nine worlds.

FIASA 2025 p49; registered review_capacity_scenarios.yaml and review_supply_worlds.yaml. Illustrative 2036 sensitivities, not calibrated production forecasts. Other plants held at 2025 reported values; Secunda gas/MRG and PetroSA restart effects remain unquantified. Low is not a Natref closure prediction.

Emerging gateways could redirect inland growth; delivered cost and reliable access determine capture

Competing route	What changes the choice	Investment implication
Durban–Lesedi	Pipeline, rail and road costs; reliable receipt and inland dispatch.	Test access and handling improvements that protect competitive customer supply.
Maputo / Matola	Delivered cost, border time and capacity into each inland destination.	Include as an emerging alternative; test customer exposure and corridor responses.
Walvis Bay	Distance, transit, border costs and feasible destination coverage.	Consider emerging inland opportunities; compare reach customer by customer.
Domestic sources	Secunda / Natref availability, product output and distribution access.	Test inland handling and connections as well as import infrastructure.

Compare common product, destination, price date and tax basis. Emerging gateways must be considered; no relative cost ranking is asserted without quotes and verified access. Landlocked SADC opportunities require separate market/access evidence.

Durban and Lesedi market sizing and investment issue tree

1 NATIONAL DEMAND

National sales baseline

2023: 21.94bn litres

2024: 20.76bn, unverified

2025 sales unavailable

Demand by use

Road, mining, industry, agriculture
Power and other uses

Demand parameters

GDP and sector activity
EVs, fleet turnover, ICE efficiency
Freight shift, prices, power dispatch

2 SUPPLY AND DELIVERY

Gross imports

By product and entry point
Exports shown separately

Ports and inland delivery network

Durban, Richards Bay, other gateways
Pipeline, road and rail to customer catchments
NMPP / TM2 / DJP links to verify

Supply and route parameters

Refinery closures / restarts and gas availability
Delivered cost, competing routes, access and contracts
Port, pipeline and rail capacity and reliability

Domestic production

Refinery location, output, yields
Own storage and distribution

3 VOPAK OPPORTUNITY

Durban accessible market

Feasible annual product volumes through
Vopak Durban to reachable customers

Lesedi accessible market

Inland demand Lesedi can feasibly serve
Catchment and receipt / dispatch connections

Capture, market share and headroom

Client throughput / accessible market = share
Accessible market less current capture = headroom
Track shared Durban-Lesedi flows once

Achievable additional capture

Customers, contracts, competition
Test capture within the opportunity

Existing assets and improvements

Working tanks, turns and constraints
Test receipt, storage and dispatch

New investment and returns

Incremental volume, cost and capex
Compare alternatives and downside

Resolution

Vopak share at each site
Scope for new investment

8 October PS framework; workbook History Q71/R71. 2023 Department sales 21.94bn L; 2024 FIASA 20.76bn L, unverified. Estimate accessible markets, then verify actual capture. Shared flows and client operating/economic inputs remain open.

Choose the investment that removes the proven constraint—and earns a return

Investment option	When it becomes relevant	Decision / commitment trigger
 Existing assets / debottlenecking	Customer growth can be served through better tank turns, scheduling or receipt/dispatch.	Use available working capacity; invest in the specific bottleneck where incremental value is positive.
 Corridor / rail access	Competitive customer flows are constrained by transport capacity, cost or reliability.	Secure service and volume commitments. Compare access contracts, loading assets and partnership investment.
 Additional tank capacity	Profitable captured flows exceed achievable handling and storage remains the binding constraint.	Size and stage tanks against contracted demand; pass the four tests and downside cases.
 EV-related adjacencies	Identified charging / fleet-energy customers offer a viable opportunity as transport electrifies.	Screen partnerships or dedicated assets separately: demand, power access, strategic fit and return must be proven.

Investment menu, not approved projects. Network rail recovery is an external fuel-demand driver; Vopak rail loading/access is a distinct investment option. EV uptake reduces some fuel demand; it does not itself prove an EV investment case. Project costs, contracts and returns remain open. Three fuel-logistics packages are priced illustratively on p39; these do not constitute engineered rail or EV project estimates.

Resolve customer volume, working capacity and project economics first to determine the investment

Outstanding request / coordinator	Required data / handback	Decision supported
DR01 Manish; Nigel review	Product/year sales, gross imports, exports, actual production and stocks. Resolve 2024 source; 2025 sales are absent.	Reconciled current market scale.
DR04 Manish; Nigel review	Port/pipeline capacity, route access and destination-matched delivered economics.	Accessible Durban and Lesedi markets.
DR07/08 Manish; Nigel/Henry review	Reported Eskom litres, fleet/efficiency, IPP timing and all refineries including PetroSA/Mossgas.	Demand and domestic-output paths.
DR02/05 Nigel coordinates	36 months of site/product movements, shared transfers, destinations, customer commitments and contracts.	Actual market share and achievable capture.
DR03 Nigel coordinates	Working/committed tankage, turns, occupancy, downtime and receipt/dispatch limits.	Operating headroom and bottlenecks.
DR06 Nigel coordinates	Storage/handling terms, costs, capex, timing, tax, working capital and hurdle rate.	Incremental returns and investment scope.

DR01-DR09 register and 8 October PS handback. Prepared, not sent; receipt not recorded. DR09 EV adjacency remains separate. Public estimates can proceed while client data is pending.

Appendix

Supporting evidence and calibration

Durban–Lesedi corridor into Gauteng

8 October 2026

Detailed evidence, assumptions and remaining calibration requirements

Photo source: Royal Vopak | Terminal Durban | vopak.com



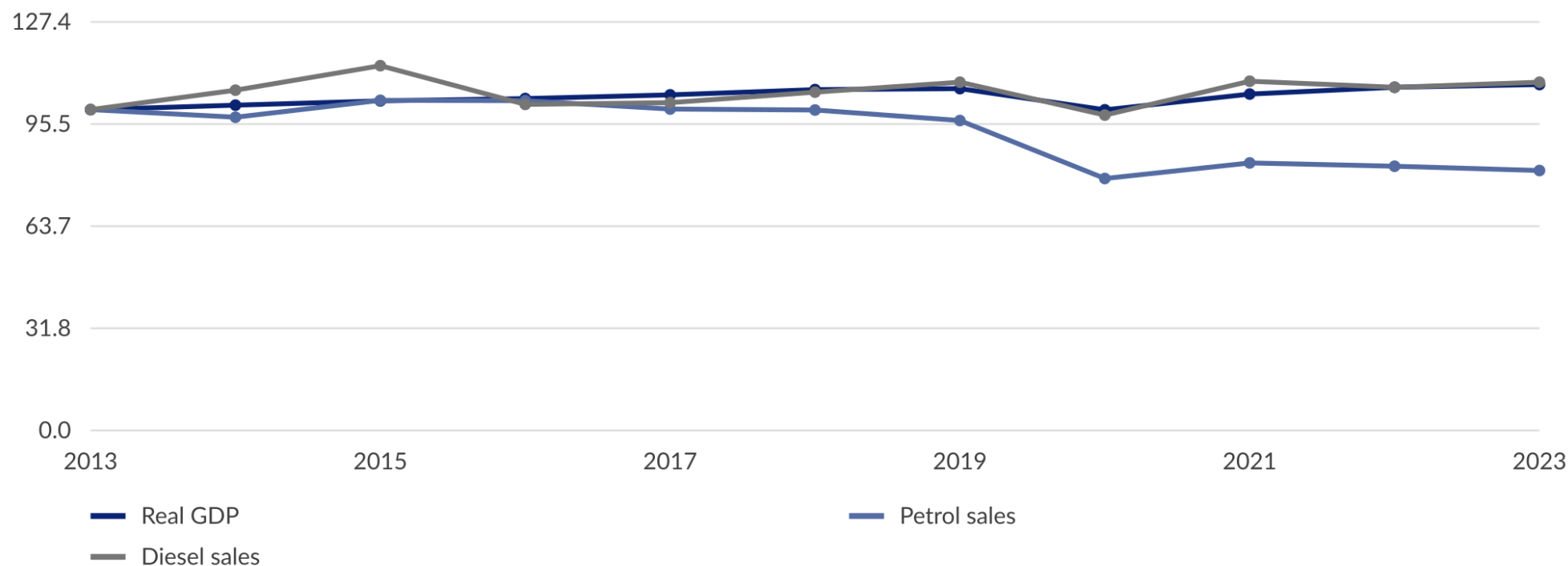
The supporting evidence calibrates demand, supply and the terminal investment tests

Evidence group	Detail retained	Connection to the story
Demand	Workbook sector history, vehicle stock/sales, diesel-use estimates, freight and power fleet.	Observed history and labelled estimates inform D1-D5; annual L/M/H paths still require agreement.
Supply	Plant output, gas availability and restart timing	S1–S4 create the three domestic-supply paths.
Capture and operations	Routes, customer share, working tankage and turnover	T1–T5 translate each world into terminal flows and constraints.
Returns	Rates, costs, capex and funding; separate EV adjacency economics.	F1 translates each investment option into cash flow and returns.

The main story runs from the opening thesis to the investment tests. The added lever register and power/supply calibration pages record what remains to be populated.

Fuel demand has diverged from GDP, so sector activity and fuel intensity need separate treatment

index: 2013 = 100; matched 2013-2023 window

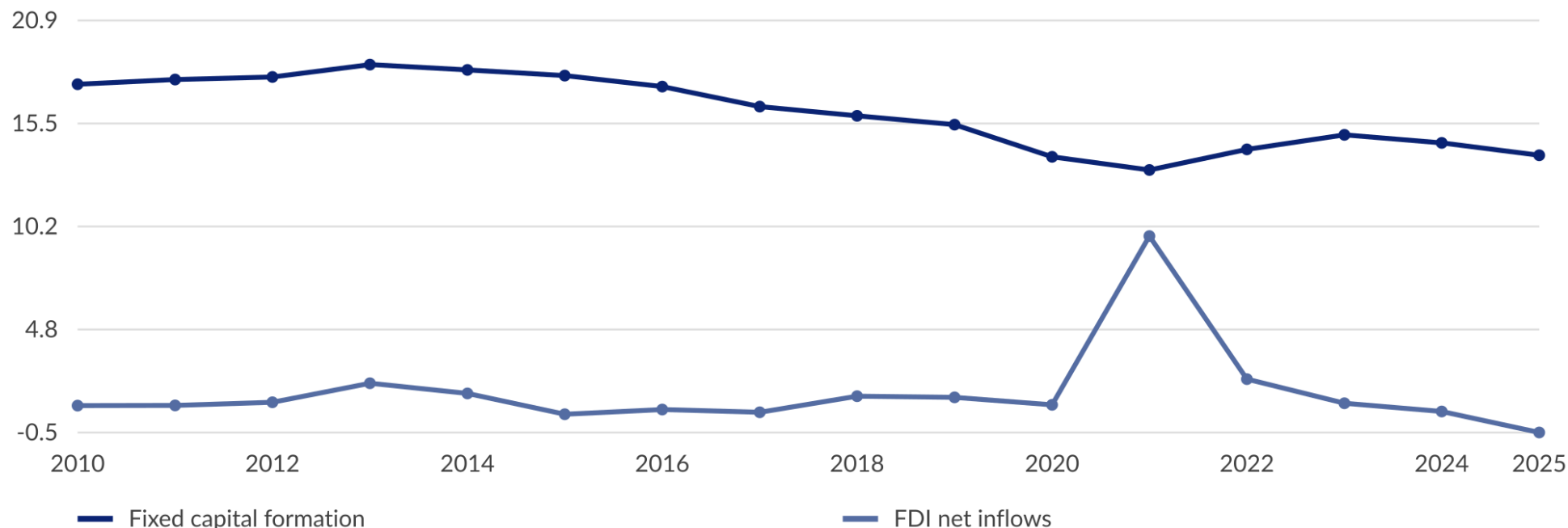


The workbook separates mining/manufacturing indices and agricultural GVA from fuel intensity. Post-2021 sector litres are estimates; GDP alone cannot explain fuel demand.

Staged GDP and department fuel sales, indices 2013=100. Sector attribution remains open; COVID distorts short-period comparisons.

Fixed investment has weakened while FDI fluctuates, limiting a simple growth-led fuel argument

percent of GDP / calendar year



Fixed investment helps frame productive capacity. FDI can include large transactions; neither series is a direct investor-confidence measure or an additional fuel-demand multiplier.

World Bank WDI, FDI net inflows and gross fixed capital formation as % of GDP, 2010–2025. Confidence series remains an evidence gap.

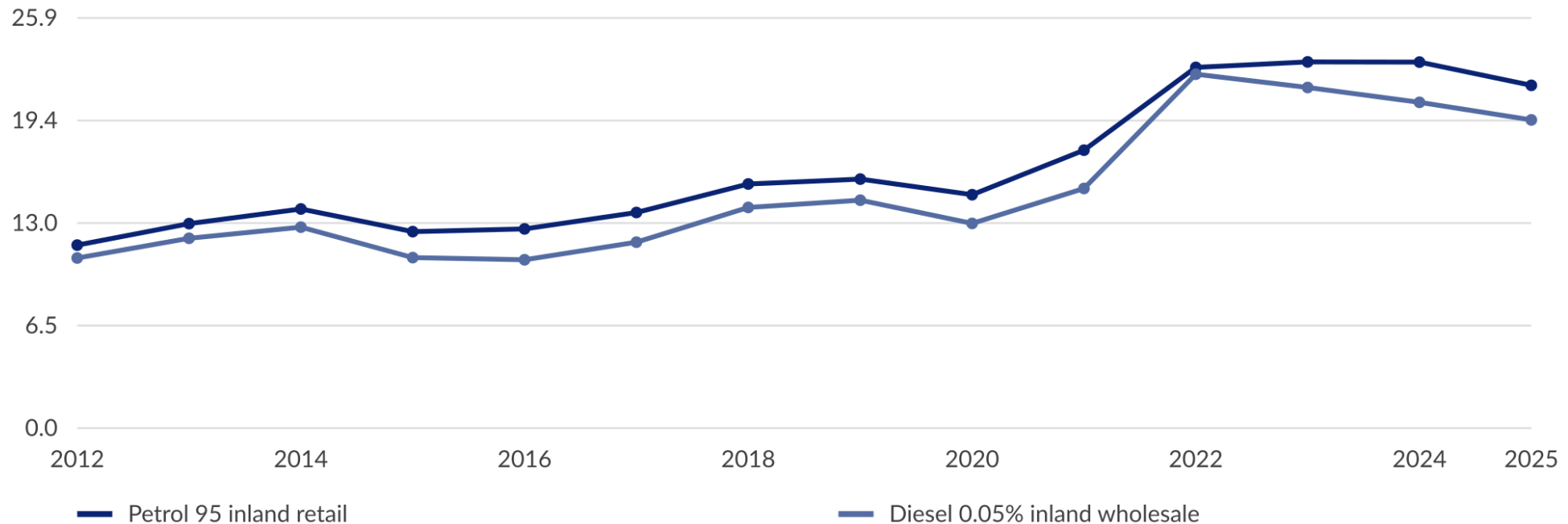
Mining, agriculture and aviation expose fuel demand to different activity drivers

Evidence	Interpretation	Implication for Vopak
Mining	2024 production index 93.5; indicative diesel 1.40bn L.	Use activity and intensity separately; balance intensity varies materially.
Manufacturing	2024 output index 93.3; indicative industry residual 0.17bn L.	Industry less mining/construction. This is not a complete industrial diesel census.
Agriculture	2025 real GVA R134.8bn (2015 prices); indicative diesel 1.13bn L.	Fuel estimate holds 2018-2021 intensity; crop and irrigation exposure remain open.
Aviation	2024 jet sales 1.96bn L; 456,214 ACSA movements; 4,285 L/movement.	FIASA sales unverified; movements omit non-ACSA airports. Fuel mix and routes are not separated.

Workbook Sector history R7/R9/R17/R19/S23/S25 and History R117/R124/R125. Sector litres after 2021 are estimates; jet 2024 uses unverified FIASA. Index base 2019=100; GVA in 2015 prices.

Higher fuel prices strengthen EV running-cost incentives, but ownership economics determine uptake

nominal ZAR/litre; full-year monthly averages



Shipping disruption, crude prices and FX can move fuel costs temporarily. EV adoption depends on sustained total ownership cost, purchase affordability, charging access and fleet replacement.

Department fuel-price history, nominal R/litre. No quantified geopolitical premium or calibrated EV adoption response is asserted.

New-vehicle efficiency gains affect fuel demand gradually as older vehicles leave the fleet

Evidence	Interpretation	Implication for Vopak
Fleet growth	2025 cars 8.18m; LCVs 2.78m; trucks 0.405m.	Registered stock shows scale; fuel within class and opening cohort age remain unknown.
Replacement	2025 apparent retirement: cars 2.36%, LCVs 3.42%. Model settings: 4% and 5%.	Stock plus sales less next stock is a diagnostic, including deregistration, exports and classification effects.
Efficiency / EVs	Study mileage/fuel-use benchmarks and separate BEV, PHEV and hybrid sales are available.	Recent OEM efficiency history remains open. Apply gains to new cohorts and count EV transitions once.

Workbook Vehicle history S6:S8/S24/S26. Apparent retirements are not verified scrappage. Stone et al. study uses a 2014 base; no recent OEM efficiency history or fuel-by-class stock is established.

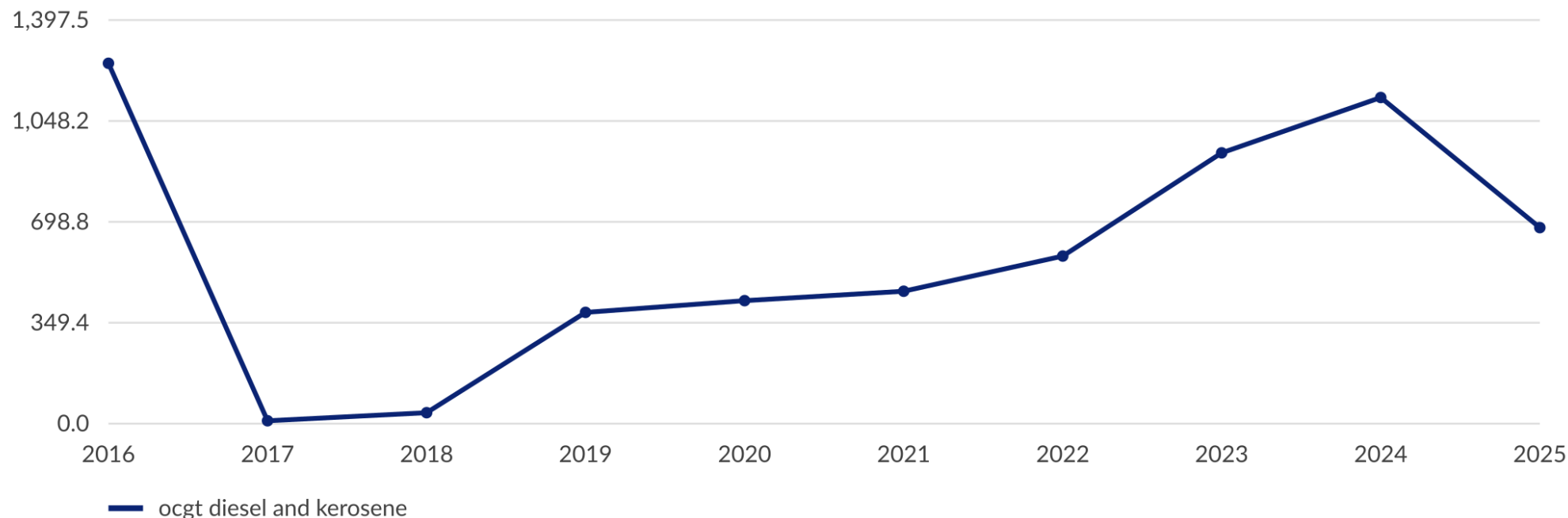
Vehicle stock and sales expose fuel-demand calibration gaps

Calibration item	Observed evidence	Model / calibration reference	Implication / limitation
2025 registered fleet	Cars 8.18m; LCVs 2.78m; trucks 0.405m.	Opening cohorts need review.	Age and fuel within class are missing; all-class fuel counts are available only for December 2023.
New-sales mix, 2025	Cars 70.7%; LCVs 24.1%; medium/heavy 5.2%.	78% / 18% / 4%.	Observed class mix differs. Agree segmentation before adopting a replacement.
Apparent retirement, 2025	Cars 2.36%; LCVs 3.42%.	4% cars; 5% LCVs.	Derived from stock and new sales; includes classification changes, exports and deregistrations.
New drivetrains, 2025	BEV 1,088; PHEV 2,810; conventional hybrid 12,818.	BEV observed share 0.18%.	All new-sales denominator. Fleet share, charging behaviour and hybrid fuel savings differ.
Petrol-use diagnostic, 2023	1,056 L per registered petrol vehicle.	Passenger setting implies 1,615 L/year.	Aggregate versus passenger scope differs. Investigate mileage and fuel-use calibration, not a direct replacement.

Workbook Vehicle history S6:S8/S18:S20/S24/S26/S31:S36/Q52. NaTIS and naamsa classes are not identical. Apparent retirement is not verified scrappage; petrol diagnostic compares different scopes.

Eskom's own OCGT fuel use fell 40% in FY2025 as availability improved

million litres / FY ending 31 March

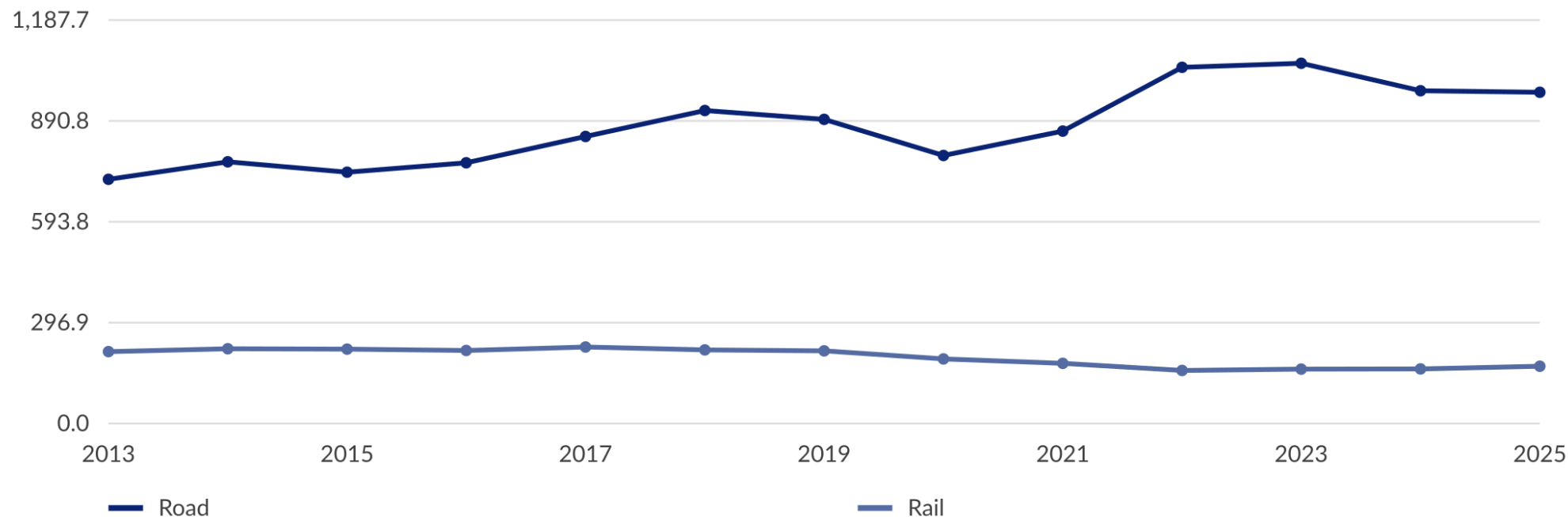


Reported own-fleet fuel fell from 1,129.5 to 679.1m L as EAF improved. Eskom/IPP output and generation-derived litres are a broader, separate series.

Eskom Integrated Report 2025, FY March, own diesel + kerosene. Workbook combined generation includes IPPs and kerosene stations; multiplying it by 0.31 L/kWh is an estimate, not reported diesel burn. Private backup remains unmeasured.

Road carries most reported freight tonnage, making transferable haul distance a key diesel lever

million tonnes / complete calendar year



2023 freight-operator fuel purchases imply ~3.15bn L at list diesel prices. Partial coverage and the estimated vehicle split do not establish corridor tonne-km.

Stats SA payload and transport-industry fuel purchases. Workbook Diesel by use Q28: spending / average diesel price, hire-and-reward firms only; may include petrol/lubricants. Road residual includes unseparated uses; tonne-km and rail traction remain open.

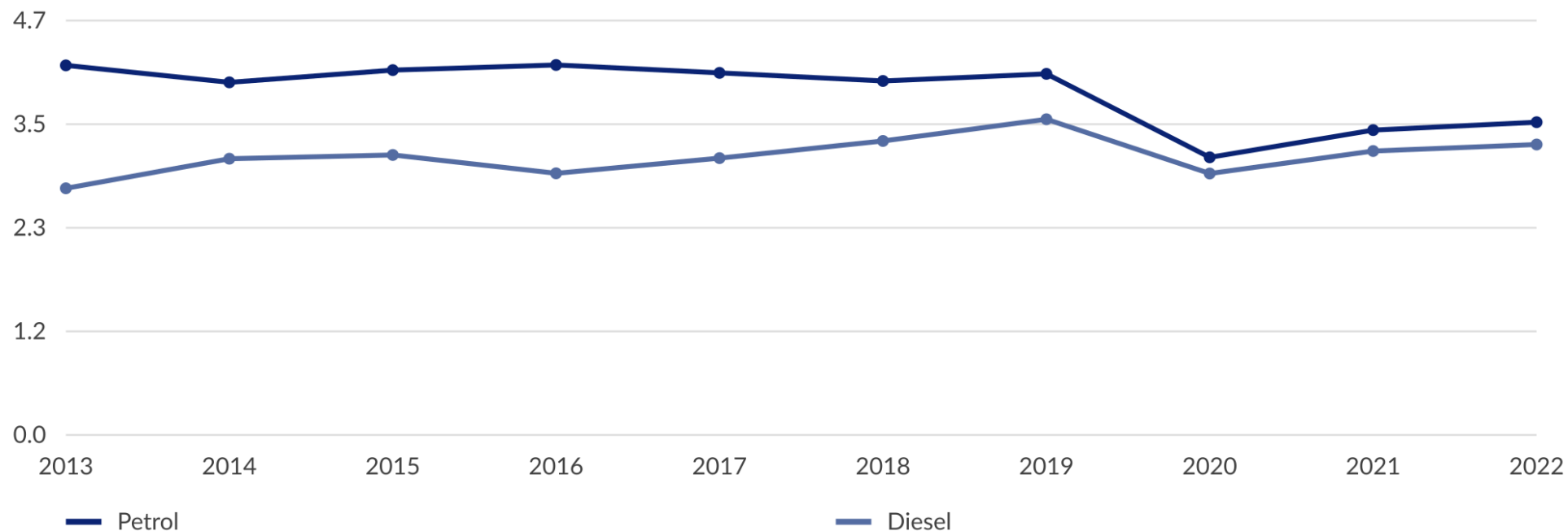
Rail funding and reliable service determine how much road freight can return to rail

Evidence	Interpretation	Implication for Vopak
Network ambition	The supplied brief reports a 250 Mt target dependent on a funding plan.	Do not treat every additional rail tonne as road displacement. Commodity, corridor and origin matter.
Funding constraint	Brief: R46.34bn approved over five years, including R1.71bn expansion.	Maintenance and sustaining spend dominate. A published plan is insufficient evidence of delivered capacity.
Container corridor	Brief: R2.11bn in 2026/27–2028/29; private funding expected thereafter.	Prioritise Durban–Gauteng road/rail competition and reconcile the corridor to Network Statement v4.
Two fuel effects	Rail can displace truck diesel while improved infrastructure also changes fuel-distribution economics.	Keep freight-demand effects separate from the route delivering fuel to Gauteng.

User-supplied AIA Transnet Funding Brief, Sep 2026, pp1–2; cites FY2026 AFS and Network Statement v4 Annexure 17. Primary figures not independently reconciled.

Gauteng demand establishes market scale, while Lesedi capture and turnover determine throughput

billion litres / complete calendar year



Gauteng sales: 6.81bn L observed in 2022; 6.81bn estimated in 2023 and 6.52bn in 2024. Customer catchment and actual Lesedi capture remain unverified.

Department provincial sales through 2022; workbook History Q84/Q104 and R84/R104 estimate later volumes. 2024 also uses unverified FIASA national sales. The chart retains observed 2013-2022 sales history, not current market share.

Every market world passes through explicit demand, supply, capture and operating levers

Lever IDs	Inputs to calibrate	Output / remaining evidence
D1–D5 demand	D1 sector growth; D2 rail; D3 EV adoption; D4 efficiency; D5 power.	29 proposed levers (20 original, nine added). 2030/2035 values need review; no agreed annual path yet.
S1–S4 supply	S1 SAPREF; S2 Secunda gas/MRG; S3 Natref; S4 PetroSA.	Output, yields and feedstock dependencies remain open. Reconcile proposed capacity, utilisation and gas assumptions.
T1–T2 capture	T1 delivered route economics/access; T2 customer/product/destination capture.	Allocate domestic and imported fuel to unique terminal deliveries. Tariffs, rights and commercial allocation remain open.
T3–T5 operations	T3 eligible working tankage; T4 turnover; T5 receipt/dispatch, peaks and stock policy.	Compare required with achievable turns, occupancy and headroom. Existing 1/2/3-turn sensitivity is illustrative only.
F1 finance	Storage/handling terms, costs, capex, tax, working capital and hurdle rate.	Translate constrained throughput into cash flow, NPV, IRR and investment triggers. Commercial inputs remain open.

Workbook HML response has 29 levers. Examples: BEV adoption, rail diversion, mileage, prices, hybrids, truck/LCV electrification and neighbouring exports. All replacement ranges remain proposals; no forecast inputs are adopted in this pack.

Generation timing must close the power balance before residual diesel can be forecast

Time series	Assumption / lever	Evidence status and next use
Recovery history	Own Eskom fuel/EAF and Eskom/IPP GWh are assembled; generation-based litres are estimated.	Match financial and calendar periods. Obtain reported litres and private-backup evidence.
Fleet and retirements	Four diesel stations: 3,089 MW. Avon/Dedisa PPA endings and coal retirement dates are assembled.	Contract end is not confirmed shutdown. Siting, renewal and actual coal decisions remain open.
Replacement / gas	Ankerlig/Gourikwa gas switch and coal-site turbines are case dependencies, not committed outcomes.	Wind/solar/gas commissioning, fuel availability and dependable output constrain peaking dispatch.
Proposed L/M/H cases	2030 power diesel: 86 / 637 / 4,224m L in the workbook scenarios.	Nigel/Henry review D22: conversion rate, backup share, load factors, repowering location and gas supply.

Workbook Power fleet F11:S15 and N63:N65; proposed scenarios, not forecasts. Generation includes kerosene. IPP renewal, gas availability and coal-site repowering are not established. Historical burn must match scope and period.

The diesel fleet and replacement timing define the power-demand cases

OPERATING DIESEL FLEET: 3,089 MW

Diesel station	MW	Owner
Ankerlig	1,338	Eskom
Gourikwa	746	Eskom
Avon	670	Independent
Dedisa	335	Independent

TIMING AND DEPENDENCIES

Avon / Dedisa PPAs end in 2031 / 2030.

Five coal sites have exemptions to March 2030; final outcomes remain open.

Gas-switch and turbine locations are assumptions, not committed projects.

PROPOSED WORKBOOK CASES – NIGEL / HENRY REVIEW

Proposed case	2030 m L	2035 m L	Key dependency
Low	86	20	Gas switch; PPAs end; no coal-site turbines.
Medium	637	637	3 GW at coal sites; gas available; diesel backup.
High	4,224	4,224	6 GW at coal sites; no gas; diesel peakers.

Gas supply, diesel intensity, backup use and coal-site repowering remain assumptions. No engine or approved scenario change is implied.

Workbook Power fleet F11:F15, N63:N65 and S63:S65. Proposed cases, Nigel/Henry review D22. 0.31 L/kWh, backup fraction, load factors, gas supply and coal-site repowering need review. Kerosene stations and private backup are separate.

SAPREF and Natref define capacity sensitivities; plant output must complete the supply paths

Plant lever	Required time-series inputs	Scenario treatment
S1 SAPREF	High capacity illustration adds 400 kbpd from assumed 2033 availability.	Preserve the hypothetical case; actual funding, schedule, ramp-up and product yields need evidence.
S3 Natref	108 kbpd remains in Medium/High; excluded in the Low 2036 stress case.	Availability stress only. No closure decision or date is established.
S2 Secunda	Operator all-products output: 29.1m barrels in FY2024 and 27.6m in FY2025.	June fiscal years; product mix differs. Gas/MRG affects liquid output; do not equate barrels with diesel litres.
S4 PetroSA	No restart addition in the existing capacity illustration; workbook identifies the source gap.	Complete the Mossgas/PetroSA feedstock, funded scope, timing and yield evidence.

Existing capacity illustrations retained. Workbook History R53/S53: Sasol all-refined-products output, June FY. Department product production stops at 2021; stocks, plant yields and restart calibration remain open.

Public capacity ratios establish scale but cannot substitute for realised terminal throughput

Catchment	Initial inference	What the result can support
Durban	45.1% of two operators' published gross Durban tankage.	Incomplete, mixed-product capacity denominator. Estimate feasible product/customer flows before calculating throughput share.
Gauteng / Lesedi	2022 sales 6.81bn L; workbook estimates 6.81bn in 2023 and 6.52bn in 2024.	Provincial sales are a demand proxy. Lesedi catchment, eligible products and actual deliveries remain unverified.
Investment implication	Accessible market less current capture gives headroom; achievable additional capture must be tested.	Track shared Durban-Lesedi flows once. Compare customer volumes with working capacity, turns and dispatch limits.

Existing capacity-scale ratios retained, not throughput shares. Workbook provincial estimates: 2023 estimated shares; 2024 estimated shares plus unverified national sales. Client throughput and commercial access remain open.

A matched demand snapshot is available, but the domestic supply balance still needs reconciliation

Evidence / bn litres	Petrol	Diesel
2023 Department sales	9.04	12.91
2024 FIASA sales, unverified	9.03	11.73
2025 SARS imports / exports	4.45 / 0.80	12.25 / 0.75
2025 sales / product output	Sales absent; product production series ends 2021.	Sales absent; product production series ends 2021.
RAF levy cross-check	Combined levied volume 24.42bn L, year to March 2025.	Fiscal/accrual versus calendar sales. Reconcile coverage before interpreting the difference.
Decision consequence	Sales less net imports remains a residual.	Stocks and definitions remain unresolved.

Workbook History Q59/Q64/R59/R64/S36:S40/R70. RAF-derived 24.42bn L is combined fuel, fiscal/accrual basis, not an adopted calendar sales total. Product output after 2021 and stocks remain missing; 2025 trade is not 2025 consumption.

The six-step bridge is implemented for screening; calibration now depends on the requested inputs

Work package	Implemented now	Input needed to complete the decision
1 Baseline	Workbook sales/trade, sector, vehicle and power history plus labelled estimates are assembled.	Agree baseline year/source; reconcile production, stocks, scopes and fiscal/calendar periods.
2 Route competition	Delivered cost and bearable-charge calculation; route readiness register.	Matched quotes, service capacity and access.
3 Vopak capture	Capture calculation constrained by accessible market and route capacity.	Customer allocation and unique terminal deliveries.
4 Existing assets	Required-turn and operating-bottleneck calculation; gross-turn screen.	Working capacity, achievable turns and peak limits.
5 Investment options	Pre-tax break-even volume calculation.	Commercial inputs and full incremental cash flows.
6 Nine worlds	Nine-cell output register and demand/supply balance calculation.	29 proposed levers and fleet cases inform review. Agree annual paths before rerunning routes, capture and returns.

Screening functions tested with synthetic cases; commercial outputs remain unpopulated. Not integrated into the forecast engine or independently reviewed. Owners, request IDs and expiry triggers: workstreams/WS0_governance/workplan/sa_investment_bridge_2026_10_08.csv.

Demand, supply and corridor access translate into captured throughput and required tank turns

Argument	Situation	Complication	Resolution / decision
 Transport demand	2023 petrol/diesel sales total 21.94bn L; sector activity and fleet evidence explain different fuel exposures.	Sector growth, rail recovery, EVs and efficiency change fuel demand.	D1–D4 define activity, rail, EV and efficiency paths. Translate them into capturable product demand.
 Power demand	Power shortages supported diesel use; Eskom consumption has recently fallen.	Recovery may persist or reverse through outages and replacement delays.	D5 adds fleet recovery, replacement timing and diesel dispatch to the same demand paths.
 Domestic supply	Reduced local production has increased reliance on imports.	SAPREF, Natref and Secunda can change domestic fuel output materially.	S1–S4 define plant output and sourcing mix. Imports are one consequence of the domestic balance.
 Corridor competition	Durban connects imported fuel to inland customers.	Alternative gateways, delivered costs, reliability and access alter flows.	T1–T2 compare delivered economics and capture for domestic and imported fuel serving common destinations.
 Vopak investment	Durban and Lesedi provide connected assets whose throughput can rise through higher turnover.	Working tankage, receipt/dispatch, peaks and stock requirements constrain achievable turnover.	T3–T5 compare working capacity, required/achievable turns and dispatch limits. F1 tests incremental returns.

Lever IDs are mapped in the appendix. All nine market worlds use explicit routing, capture and operational assumptions. Tank occupancy, turnover and throughput utilisation are separate measures. Competitiveness varies by destination; neither alternative gateway is yet established as competitive into Gauteng.

The illustration gives extra tanks only R8m more NPV than improvement alone

High demand / low supply	Capital and throughput	Incremental value versus existing
Existing	R0m capex 1.890bn L/year served	NPV R+0.0m Unserved 0.923bn L/year
Improve access / dispatch	R100m capex 2.268bn L/year served	NPV R+27.0m Unserved 0.545bn L/year
Improve + add tanks	R600m capex 2.813bn L/year served	NPV R+35.1m Unserved 0.000bn L/year
Decision threshold	Improvement beats existing above 2.211bn L/year.	Tanks beat improvement above 2.806bn L/year; assumed capture 2.813bn.

ILLUSTRATIVE ONLY. Combined package = R100m access/dispatch improvement + R500m for a further 40,000 gross m³ beyond the reported 140,000 m³. Alternatives are authored packages, not engineered designs. Full incremental pre-tax cash flows in delivered CSV; 100% site basis. NPV uses 10% and 20 operating years; commercial and delivery risks uncalibrated.

Vopak reports anchor capacity and the revenue model; they do not disclose Lesedi project economics

Evidence class	What the sources establish	How it is used
REPORTED Assets	Lesedi: 140,000 m ³ ; Durban: 360,246 m ³ . Vopak share: 70% at each site.	Full physical capacity for operations. The commissioned 40,000 m ³ Lesedi expansion is included in 140,000 m ³ .
REPORTED Revenue	Storage services: EUR1,058.8m of EUR1,298.9m consolidated revenue in 2025.	81.5% is storage revenue. Rental may include minimum throughput; extra turns are not automatically extra revenue.
INFERRED Group ratio	EUR93.9 revenue / gross m ³ /year: proportional revenue divided by year-end proportional capacity.	Mixed-portfolio scale indicator only. Not a Lesedi tariff, margin or estimate of site revenue.
REPORTED Benchmarks	15.6% group operating cash return; 5–7x capex/EBITDA for gas/industrial and 4–8x for transition infrastructure.	Context, not South Africa fuel-project hurdle rates. Authored 10% discount rate is separate.

Vopak Annual Report 2025 pp36/39, 63, 213 and 324; FY2025 Analyst Presentation p24; official terminal directory checked 8 Oct 2026. All other business units includes multiple countries; do not allocate its profit to South Africa. URLs, downloaded originals and hashes retained.

The investment result rests on explicit assumptions that must be replaced or agreed

ILLUSTRATIVE input	Value used	What must be validated
Catchment balance	Demand 6.0 / 6.8 / 7.6bn L; domestic deliveries 2.4 / 3.6 / 4.8bn L.	Steady-state stress points; M rounds 2022 Gauteng sales for scale. No national production forecast implied.
Capture	70% of imports via Durban; Vopak captures 80%; 90% goes to Lesedi; 8% domestic capture.	No observed shares. Capture is unique Lesedi flow; Durban transfers are not added to market demand.
Operations	90% working fraction; turns 1.25 existing / 1.50 improved; +40,000 gross m ³ tanks.	126,000 working m ³ initially; 162,000 with tanks. Assumed receipt/dispatch ceiling 3.5bn L/year.
Revenue	R1,500 / gross m ³ /year rent on added contracted tanks; R60 / extra m ³ contribution.	Chargeable excess services only, excluding rent/minimum-throughput inclusions. Added rent follows incremental storage use.
Capital and costs	R100m improvement + R500m tanks; fixed opex R5m + R12m/year; maintenance 1% capex/year.	Author-selected cost sensitivities; no Vopak quotation or engineering estimate.
Cash flow	20 years at 10%; 50% / 75% / 100% operating ramp; capex upfront.	Pre-tax unlevered constant ZAR. Tax, working capital, construction timing and residual value excluded.

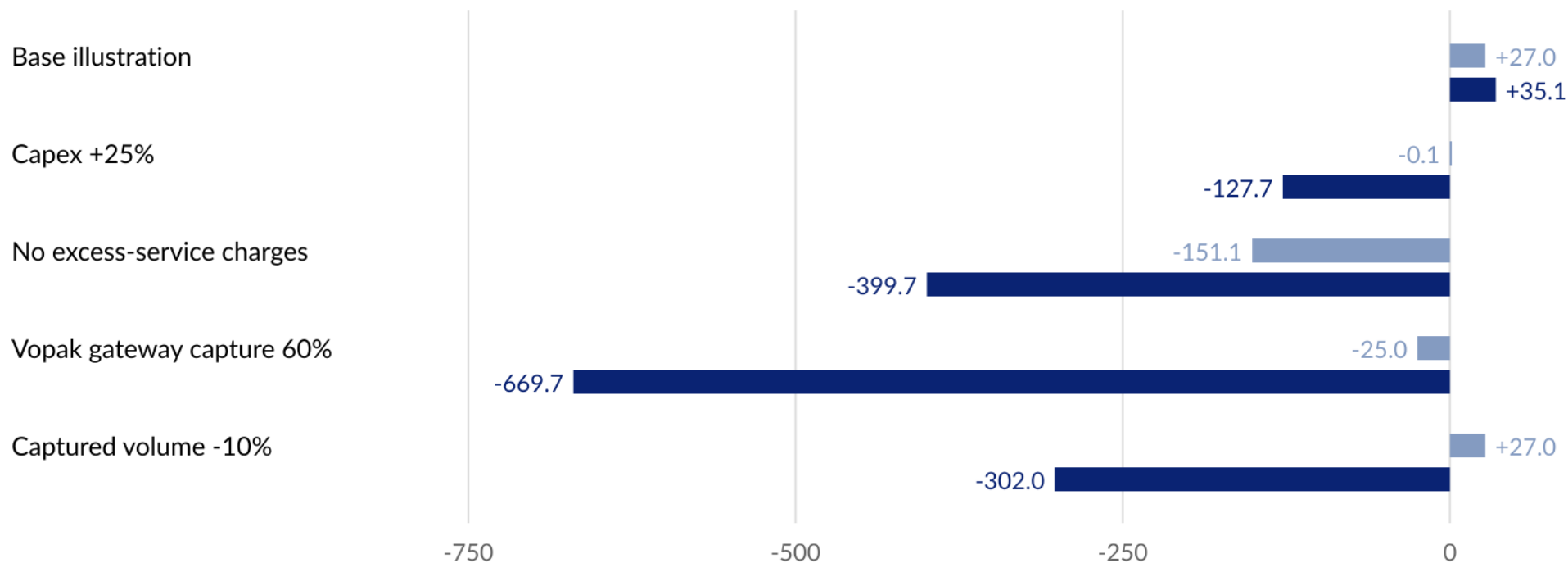
Every illustrative value is registered in assumptions/2026/vopak_investment_illustration.yaml. Not a forecast, commercial offer or approval. Proposed validation: Nigel coordinates Vopak commercial/operations; Manish reconciles; independent review outstanding.

Small changes in capture or economics overturn the illustrated tank-expansion choice

ILLUSTRATIVE incremental NPV, Rm

Improvement

Improvement + tanks



The illustration supports validating contracts and a low-capex improvement first. Extra tanks have only a narrow advantage and fail the tested downside cases.

ILLUSTRATIVE high-demand/low-domestic-delivery case. NPVs are incremental versus existing assets at 10%, pre-tax. Each shock applied separately: capex +25%; no chargeable excess services; Vopak gateway capture 80%→60%; captured Lesedi flow -10%. No probabilities. Full numeric outputs and cash flows retained.

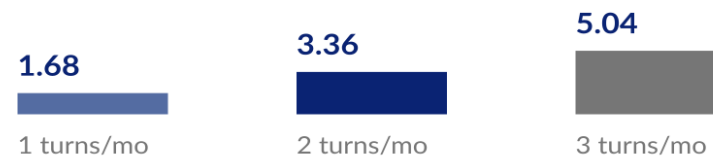
Illustrative tank turns test handling potential; actual operating limits still need validation

Gross handling sensitivity

Durban



Lesedi



bn litres/year; 1 / 2 / 3 monthly turns are an existing authored sensitivity, not measured site limits.

Investment test

01 | Calculate required turns

Captured annual throughput divided by eligible working capacity, then by 12.

02 | Test achievable handling

Check receipt / dispatch, pipeline access, seasonal peaks and stock policy.

03 | Identify the binding constraint

Use existing assets first; debottleneck or add tanks where the constraint is profitable.

Next action

Confirm working tankage and achieved turns before sizing any additional storage.

Required working-capacity turns must be compared with achievable turns at each terminal. Gross handling equivalents do not establish spare capacity or actual market share.

Registered REG-TERMINAL-HANDLING-TURNS: 1/2/3 turns per month, authored operational sensitivity, not the demand-supply worlds or measured limits. Gross Durban mixed-product tankage; no assumed eligible fraction. Do not sum inter-terminal transfers as unique demand. Separate p4/p5 illustration assumes 90% working capacity and 1.25/1.50 turns; neither is inferred from group occupancy.